# MDADet: A Multimodal Dynamic Adaptation Framework for Efficient Small Object Detection in Aerial Images

Jian Zhang, Jiarong Lv, Heng Zhang, *Member, IEEE*, Ming Li, and Meng Huang, *Member, IEEE*

***Abstract*—In this article, we propose a multimodal dynamic adaptive detection framework (MDADet) tailored for small object detection named MDADet. Concretely, we utilize a dynamic IoU-centric slicing-based data augmentation (DICSA) strategy to prioritize high-intersection over union (IoU) regions during training. The strategy effectively eliminates redundant background information and significantly accelerates model convergence. In addition, the robustly optimized BERT pretraining approach (RoBERTa) encodes bounding box annotations into semantic embedding, which are fused with image features via a transformer to generate multimodal representations for small object recognition. The knowledge distillation is utilized to transfer capabilities from the multimodal teacher model to a lightweight multimodal student model, reducing parameter scale and improving inference speed. During fine-tuning of the single-modal student model, the transformer encoder is frozen, and a lightweight feature pyramid integrated with pixel-shuffle and hierarchical detection heads is incorporated, ensuring robust performance even without textual input. Experimental results compared with other methods demonstrate the effectiveness and advancement of MDADet, achieving 81.07% mAP on DOTA 1.0, 86.76% on VEDAI, 73.55% on DIOR, and 97.61% classification accuracy on NWPU VHR-10, with a model size of only 37.8-M parameters.**

***Index Terms*—Adaptive slicing, knowledge distillation, multimodal fusion, pixel-shuffle, small object detection.**

## I. Introduction

AUTONOMOUS aerial vehicles (AAVs) have been widely deployed in various fields [1], [2]. However, aerial object detection remains challenging due to several inherent difficulties [3]: small objects with low pixel resolution lead to high miss rates; complex backgrounds with texture interference and lighting variations increase errors; the distinctive top-down perspective limits model transferability from standard datasets; and random positional and rotational variations demand robust detection methods.

Object detection frameworks are commonly categorized into single-stage and two-stage paradigms. Two-stage detectors first generate region proposals and then perform refined classification and localization, achieving strong accuracy but at the cost of higher computational overhead and latency, which limits their suitability for real-time AAV applications [4]. In contrast, single-stage detectors directly regress object categories and bounding boxes from dense predictions, offering superior inference speed, simpler pipelines, and better deployment efficiency on resource-constrained platforms. Building on these advantages, one-stage detectors such as the YOLO series [5] have become mainstream for AAV-based tasks due to their favorable balance between speed and accuracy [6]. However, they still struggle with unique aerial imaging challenges. Specifically, small objects often suffer from semantic noise, blurred edges, and poor cross-scale feature alignment [7], [8]. Dense and occluded scenes exacerbate feature confusion, and the long-tail class distribution further hampers the recognition of rare objects [9]. Moreover, traditional methods struggle with arbitrarily oriented detection and significant viewpoint variations, highlighting the need for more specialized and robust solutions [3], [10].

Recent advances in multimodal fusion and vision–language modeling have improved detection robustness, yet most existing approaches rely on RGB–thermal [11], RGB–depth [12], or RGB–audio [13] inputs that require dedicated sensors or specialized datasets, limiting their applicability in AAV scenarios. Annotation-derived textual descriptions provide a universally accessible semantic modality without additional acquisition cost, but remain underutilized in aerial detection. Meanwhile, knowledge distillation has become an effective way to transfer semantic priors from large models to lightweight detectors, though current methods focus predominantly on single-modality visual distillation. Multimodal distillation, particularly transferring text-guided semantics to image-only models, remains insufficiently investigated and often depends on external corpora or paired multimodal datasets, restricting scalability.

To address these challenges, we introduce a multimodal dynamic adaptive detection framework (MDADet) designed

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for AAV small-object scenarios. The method integrates text–image fusion through a transformer encoder and transfers multimodal semantic priors to a single-modal student model via a structured distillation pipeline, thereby improving detection robustness under complex aerial conditions. Specifically, this work: 1) develops a cross-modal fusion architecture with a teacher–student distillation mechanism that embeds high-resolution visual cues and linguistic priors into the student model; 2) it proposes a dynamic slicing-based augmentation strategy and a multiscale composite loss that jointly mitigate background redundancy and strengthen the optimization of small, low-contrast objects; and 3) it designs an efficient lightweight multiscale feature fusion module with hierarchical task allocation and fast inference acceleration. The main contributions of this work are as follows.

1) *Multimodal Fusion and Cross-Modal Adaptation Architecture:* We design a multimodal teacher model that fuses high-resolution images with robustly optimized BERT pretraining approach (RoBERTa)-encoded positional descriptions. We transfer cross-modal semantics to the student model using a three-stage knowledge distillation strategy that aligns features, enforces contrastive representations, and supervises the detection head. We perform multimodal distillation and then freeze the encoder to enable robust image-only inference, preserving the semantic benefits in real-world applications.
2) *Dynamic Data Augmentation and Multiscale Composite Loss Design:* We propose dynamic IoU-centric slicing-based data augmentation (DICSA) to reduce background noise and address the sparsity of small objects in aerial imagery. We also design a multiscale composite loss that considers object area, edge distance, center offset, and aspect ratio, enhancing the localization and classification of small, marginal, and low-contrast objects in complex aerial scenes.
3) *Lightweight Multiscale Feature Fusion:* We use convolutional layers and pixel shuffle to efficiently generate multiscale features from the Transformer output, enabling up-sampling while preserving high-resolution detail. We assign small objects to high-resolution features and large objects to low-resolution features, and employ fast NMS for parallel candidate screening, improving detection speed and efficiency.

## II. Related Works

### A. Object Detection in Drone Aerial Photography

Drone aerial photography often involves objects with arbitrary orientations, making accurate representation challenging. Early methods used rotating bounding boxes $(x, y, w, h, \theta)$ to mitigate boundary issues from angle periodicity [14], but their angle periodicity causes regression discontinuities. To solve this issue, various improved rotating bounding box representations have been proposed. For example, gliding vertex [15] and RSDet [16] use external horizontal bounding boxes and the offsets of four vertices to describe objects; oriented R-CNN [17] replaces the angle with midpoint offsets, forming a six-parameter representation $(x, y, w, h, \Delta\alpha, \Delta\beta)$, significantly improving detector performance. Although these methods avoid sudden changes in definition, they struggle to accurately describe the positions and orientations of irregularly shaped objects.

### B. Model Architectures

Early AAV aerial object detection relied on convolutional neural networks (CNNs) like AlexNet and VGG [18], which used convolution and pooling layers to automatically extract hierarchical features. However, their performance was limited by challenges in capturing contextual information and handling multiscale targets within large, complex aerial scenes.

Feature pyramid networks (FPNs) [19] address multiscale object detection in remote sensing by merging multiscale features via top-down and lateral connections, enabling size-specific detection across pyramid levels. In drone imagery, FPN variants adapt to elongated targets through structural adjustments and employ techniques like SNIPER, which selectively samples scale-specific features to enhance multiscale training efficiency and detection performance.

Recent advances in drone aerial object detection include vision transformers (ViTs) [20], capturing long-range dependencies via self-attention, Swin Transformer [21], enhancing multiscale target modeling with hierarchical window attention, and DETR-RS [22], improving rotating target detection through rotation-invariant mechanisms. Hybrid CNN-Transformer models also emerged. Meanwhile, large kernel convolutions gained prominence, with PKINet [23] using multiscale depthwise kernels for texture features and LSKNet [24] employing kernel selection mechanisms for contextual modeling, offering competitive alternatives to Transformers in aerial tasks.

Despite the effectiveness of FPN, Transformer, and large-kernel convolution architectures in AAV object detection, performance remains limited by low resolution and background noise in aerial imagery. To improve robustness in such complex scenarios, incorporating multimodal fusion offers a promising solution, as it leverages complementary information to suppress noise and enhance detection accuracy.

### C. Loss Functions

In AAV aerial target detection, loss functions are divided into two categories: classification loss and localization loss.

To address the class imbalance problem in aerial images, focal loss [25] is commonly used. It introduces a modulation factor to reduce attention to easily classified samples, thereby guiding the model to focus on hard-to-classify samples or imbalanced classes and improving overall detection accuracy. Specifically, the formula for focal loss is

$$\mathrm{FL}(p_i) = -\alpha_i (1 - p_i)^{\gamma} \log(p_i) \tag{1}$$

where $p_t$ represents the model's predicted probability for the target class, $\alpha_t$ is a balance factor (with a value range of [0, 1]) used to resolve the imbalance between the number of positive and negative samples, and $\gamma$ is a focusing parameter $\gamma \geq 0$ used to reduce the weight of easily classified samples. Localization

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loss functions [26] are used to quantify the difference between predicted bounding boxes and ground-truth bounding boxes. Specifically, the formula for the localization loss function is as follows:

$$\text{CRLoss} = \lambda_1 \cdot \text{IoULoss} + \lambda_2 \cdot \text{AngleLoss} + \lambda_3 \cdot \text{ShapeLoss}. \tag{2}$$

Here, $\lambda_1$, $\lambda_2$, and $\lambda_3$ are weight coefficients that balance each loss term and need to be adjusted according to the dataset.

Common methods such as L1 loss [27] and L2 loss [28] are prone to being affected by outliers, leading to unstable training. Smooth L1 loss achieves optimization through dynamic switching: it adopts L2 loss for small errors and L1 loss for large errors. This enhances robustness to outliers [29] while improving bounding box regression accuracy. Specifically, the standard mathematical expression for Smooth L1 Loss is

$$\text{Smooth L1}(x) = \begin{cases} \frac{1}{2}x^2 & \text{if } |x| \le 1 \\ |x| - \frac{1}{2} & \text{if } |x| > 1 \end{cases} \tag{3}$$

where $x$ is the difference between the predicted value and the ground-truth value $x = \hat{y} - y$, with $\hat{y}$ being the predicted value and $y$ being the ground-truth value.

Since AAV aerial target detection usually requires simultaneous target classification and localization [30], a multitask [31] joint loss has been widely applied.

## III. Proposed Method

This study introduces a novel multimodal learning framework for aerial object detection, which synergistically integrates textual annotations and visual information to enhance detection performance under complex scenarios. Specifically, we propose a DICSA strategy that dynamically adjusts spatial sampling regions based on intersection over union (IoU) metrics. This approach enhances training efficiency by focusing on regions with high object-context relevance while ensuring computational feasibility.

Furthermore, this work is one of the first attempts to construct a text–image multimodal detection paradigm for aerial imagery. The proposed framework incorporates a Transformer encoder-based cross-modal fusion module that jointly encodes visual features and textual priors through co-attention mechanisms, yielding semantically enriched representations. To preserve multimodal reasoning capabilities during text-absent inference stages, we develop a progressive knowledge distillation framework comprising three integrated components.

1) Adaptive object assignment and multitask optimization, which balance regression losses based on predicted IoU values.
2) Feature layer distillation, which enforces cross-modal embedding alignment via contrastive learning.
3) Detection head distillation, which preserves cross-modal knowledge through parameter regularization.

To facilitate efficient knowledge transfer, we introduce an IoU-aware dynamic weighting strategy that modulates loss contributions according to localization confidence, ensuring stable convergence across varying object scales. Addressing challenges posed by extreme scale variations and complex object configurations in aerial scenarios, we propose a lightweight feature pyramid construction (LFPC) methodology coupled with pixel-shuffle up-sampling (PSU). This architecture employs deformable convolutional layers for dynamic receptive field adjustment and channel-wise feature recombination through PSU modules, which assemble hierarchical features into spatially coherent representations while maintaining computational efficiency. The diagram is shown in Fig. 1.

### A. DICSA for AAV Imagery

Traditional fixed-window image slicing methods often introduce excessive background noise, causing category imbalance and hindering small-object feature learning [32]. To address this, we propose DICSA, an adaptive slicing framework based on dynamic IoU criteria, as illustrated in Fig. 2.

Given an aerial image $I \in \mathbb{R}^{H\times W\times 3}$ and its annotated target bounding box set

$$B = \{b_i \mid b_i = (x_i, y_i, w_i, h_i)\}_{i=1}^{N}. \tag{4}$$

*Candidate Slice Generation:* Using a sliding window with overlap, a candidate slice set $S = S_j$ is generated. For each slice $S_j$, the DICSA IoU criterion is defined as

$$\text{IoU}_{\text{DICSA}}(S_j, B_i) = \max\left(\frac{\text{Area}(S_j \cap B_i)}{\text{Area}(S_j)}, \frac{\text{Area}(S_j \cap B_i)}{\text{Area}(B_i)}\right). \tag{5}$$

*Explanation of (5):* This formula ensures that as long as a slice $S_j$ contains any portion of the target $B_i$, $\text{IoU}_{\text{DICSA}}(S_j, B_i)$ will equal 1. Specifically, if the target is fully contained within the slice, $(\text{Area}(S_j \cap B_i)/\text{Area}(B_i)) = 1$. If the slice is fully covered by the target, $(\text{Area}(S_j \cap B_i)/\text{Area}(S_j)) = 1$. The use of the max operator ensures that either condition is sufficient for $\text{IoU}_{\text{DICSA}}$ to reach 1. Thus, (5) guarantees complete retention of target information within each retained slice, while slices without sufficient target content are discarded, effectively suppressing background noise.

Slices are retained if $\text{IoU}_{\text{DICSA}}(S_j, B_i) \ge \tau$, where the adaptive threshold is defined by

$$\tau = 0.5 \cdot \left(1 + \frac{\min(w_i, h_i)}{\max(w_i, h_i)}\right). \tag{6}$$

Since $0 \le (\min(w_i, h_i)/\max(w_i, h_i)) \le 1$, $\tau$ always lies within [0.5, 1]. This adaptive threshold allows DICSA to dynamically accommodate variations in target aspect ratio and slice size without requiring threshold tuning, preserving comprehensive contextual information for small and irregularly shaped targets and promoting faster model convergence.

### B. Adaptive Object Assignment and Multitask Optimization

The proposed adaptive object assignment and multitask optimization combine IoU-based regression (AEF-IoU) with adaptive weighting to dynamically prioritize small objects. It further enhances edge detection and emphasizes difficult samples via focal loss. In addition, center and aspect ratio

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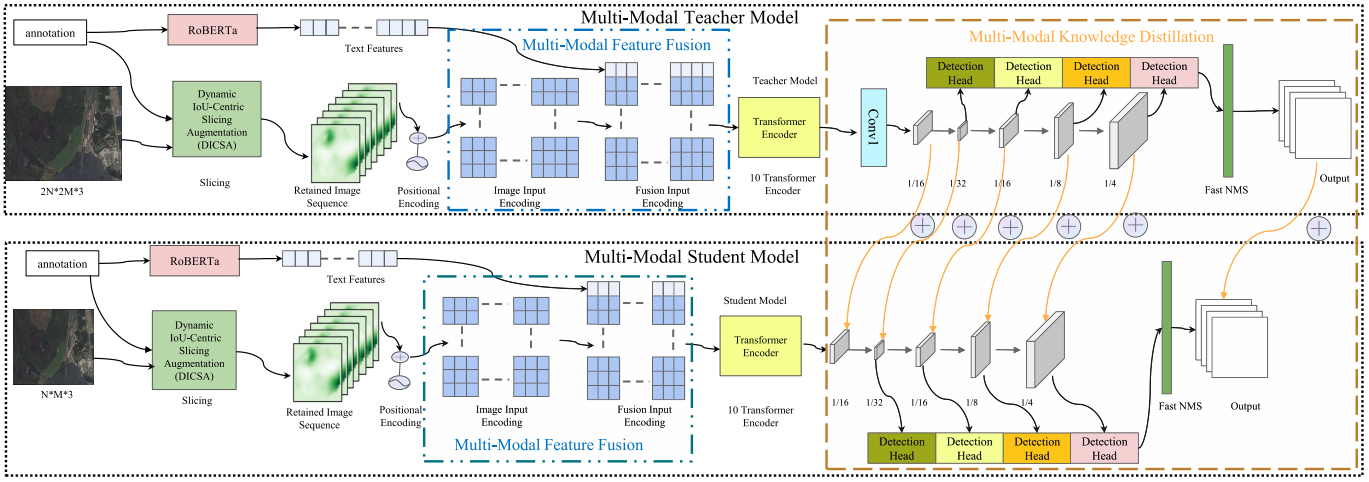



Fig. 1. Architecture of MDADet model.

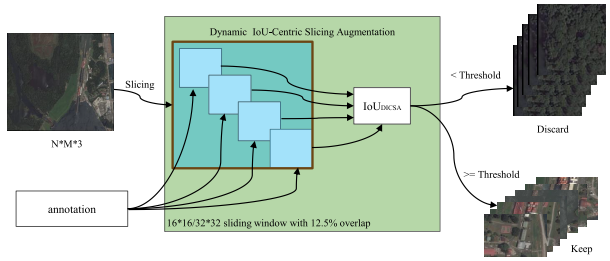



Fig. 2. Architecture of DICSA.

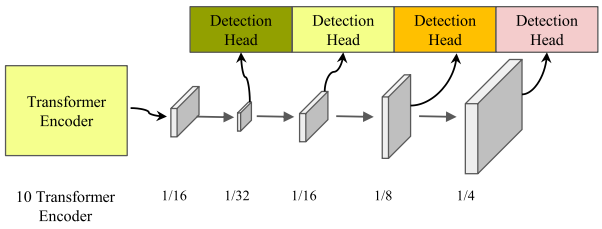



Fig. 3. Diagram of lightweight architecture design.

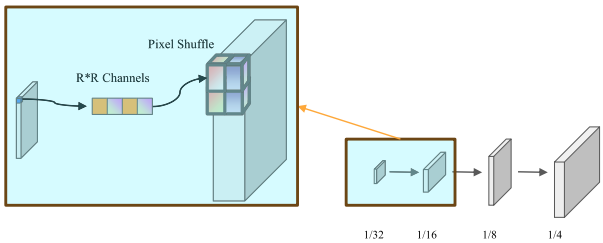



Fig. 4. Diagram of up-sampling mechanism.

constraints are introduced to stabilize bounding box predictions. Collectively, these components improve the precision and robustness of small, edge, and rotating object detection in AAV imagery. The formula is as follows:

$$L_{\text{AEF-IoU}} = \frac{-\alpha\,(1-p_t\cdot \text{IoU})^{\gamma}\log\left(p_t\cdot \text{IoU}\right)}{w\cdot h+\mathfrak{S}} \cdot \left(1-\left(\text{IoU}-\frac{d^2}{c^2}-\alpha v\right)\right) \tag{7}$$

$$d^2 = \left(\text{vx}_{\text{pred}}-\text{vx}_{\text{gt}}\right)^2+\left(\text{vy}_{\text{pred}}-\text{vy}_{\text{gt}}\right)^2 \tag{8}$$

$$v = \frac{4}{\pi^2}\left(\arctan\left(\frac{w_{\text{pred}}}{h_{\text{pred}}}\right)-\arctan\left(\frac{w_{\text{gt}}}{h_{\text{gt}}}\right)\right) \tag{9}$$

where $\alpha$ and $\gamma$ are focal loss parameters that balance class weights and emphasize hard samples. $p_t$ is the predicted probability of the correct class; simultaneous classification and regression ensure increased weighting of hard-to-classify samples. IoU measures the overlap ratio between the predicted and ground truth bounding boxes. $w * h$ denotes the ground truth box area; a small constant is added to prevent division by zero. $c$ is the diagonal length of the smallest enclosing box. $d$ denotes the center distance between the predicted and ground truth boxes, while $v$ modulates the loss via arctangent difference of their aspect ratios.

### C. Efficient Adaptive Construction of Feature Pyramid

To address feature fusion challenges from large-scale variations and complex AAV aerial objects, we propose an adaptive feature pyramid that enhances detection accuracy and efficiency [33]. This is achieved via dynamic anchor generation, multiscale optimization, lightweight pyramid design, hierarchical task allocation, and accelerated NMS, overcoming the computational redundancy and detail loss of traditional FPNs.

*1) LFPC With PSU:* To address the computational redundancy and loss of high-resolution information in traditional FPN, its key improvements include.

*Lightweight Feature Pyramid Construction:* A specialized convolutional module is designed on the 1/16-scale Transformer encoder output to down-sample both embedding and sequence dimensions efficiently (as shown in Fig. 3). Initially, a stride-4 convolution reduces embedding dimensionality for feature compactness, followed by a stride-1 convolution to preserve sequence information and finally a stride-2 convolution further down-samples to the 1/32 scale, enabling hierarchical feature representation.

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*Pixel-Shuffle Up-Sampling:* Following the generation of the 1/32 scale feature map, PSU is employed to reconstruct multiscale features (1/16, 1/8, and 1/4), progressively restoring spatial details (as shown in Fig. 4).

Pixel-shuffle reconstructs high-resolution feature maps by spatially rearranging the channel dimension of low-resolution features, which scales linearly with the input resolution as $O(H \times W \times C)$. In contrast, bilinear and nearest-neighbor interpolation operate directly on the enlarged spatial domain, leading to complexities of $O(H \times W \times r^2)$, where $r$ denotes the upsampling factor. This scaling difference becomes increasingly significant in multiscale architectures, where repeated interpolation amplifies computational redundancy. By avoiding these spatial-domain operations, pixel-shuffle reduces computation by approximately 70% compared with bilinear interpolation under the same up-sampling ratio. Furthermore, the global contextual information provided by the Transformer encoder at the 1/16 scale allows pixel-shuffle to restore spatial details efficiently while preserving semantically enriched small-object cues that interpolation-based methods frequently blur. This mechanism not only enhances the expression of fine-grained features but also provides a more efficient and structurally coherent reconstruction pathway for multiscale detection heads. The integration of pixel-shuffle therefore ensures a superior balance between computational efficiency and representational fidelity, supporting high-precision small-object detection in complex aerial imagery environments.

By expanding channel information to extract small object features as much as possible, the channel information is then assembled into a feature map in a specific order for recognition. This mechanism enhances small object feature retention and supplies high-quality inputs for multiscale heads, significantly boosting detection performance.

*2) Hierarchical Detection Task Allocation and Accelerated Inference:* Traditional fixed-anchor detectors struggle to handle the large-scale variations in AAV imagery, from nearby large to distant small objects. To address this, a dynamic anchor generation and multiscale matching mechanism is introduced, where $K$-means clustering derives anchor parameters from object size distributions, removing manual bias. This approach clarifies detection responsibilities by aligning anchor sizes with corresponding feature map resolutions. The high-resolution layer (1/4) is responsible for detecting small objects with dense anchor coverage, whereas the low-resolution layer (1/32) focuses on large object detection using a sparse anchor layout to minimize redundancy.

### D. Multimodal Knowledge Distillation and Fine-Tuning

Building on these innovations, we propose a text–image multimodal teacher model to enhance semantic representation by fusing visual information with textual priors. It transfers multimodal advantages through a three-stage knowledge distillation and employs two-stage training. This design enables high-performance, image-only detection while meeting real-time and lightweight deployment requirements.

*1) Multimodal Teacher Model Training:* The multimodal teacher model (Fig. 5) incorporates textual information to enhance object detection by integrating up-sampled images, quadrilateral bounding box coordinates, and corresponding natural-language descriptions. For each annotated object, we extract its semantic category together with the coordinates [($x1$, $y1$), ($x2$, $y2$), ($x3$, $y3$), and ($x4$, $y4$)] and convert them into structured natural-language statements following a unified template, such as: "there is a pedestrian at coordinates [($x1$, $y1$), ($x2$, $y2$), ($x3$, $y3$), and ($x4$, $y4$)]." This procedure is applied uniformly across all datasets to ensure linguistic consistency and semantic integrity. The generated descriptions are encoded using RoBERTa to obtain fixed-length text embeddings, which serve as semantic tokens complementing the visual representations. To construct the multimodal input sequence, the text embeddings are concatenated above the position-encoded visual patch embeddings, as shown in the "fusion input encoding" component in Fig. 5, where the white blocks denote textual embeddings and the blue blocks denote image embeddings. This ordered stacking enables the

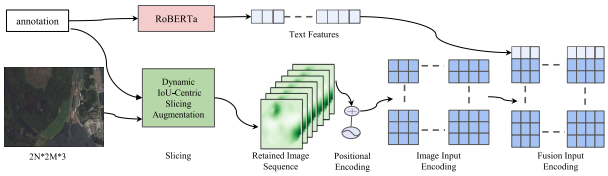



Fig. 5. Diagram of multimodal feature fusion.

**Algorithm 1** Multimodal Feature Fusion

```
object_list ← extract_annotated_objects(single-image)
text_embedding_list ←[-DEL]
for each object in object_list do
    category ← object.category
    coordinates ← object.coordinates
    description ←"There is a " + category + " at coordinates " + coordinates_to_string(coordinates)
    text_embedding ← RoBERTa_encoder(description)
    append text_embedding to text_embedding_list
end for
image_embeddings ← extract_image_patch_embeddings(input_image)
position_encoded_image_embeddings ← add_position_encoding(image_embeddings)
multimodal_sequence ←[-DEL]
for each text_embedding in text_embedding_list do
    append text_embedding to multimodal_sequence
end for
pos_img ← position_encoded_image_embeddings
for each image_embedding in pos_img do
    append image_embedding to multimodal_sequence
end for
if length(multimodal_sequence) >0 then
    transformer_input ← multimodal_sequence
    return: transformer_input
else
    return: Sequence construction failed for single image
end if
```

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Transformer to access all global textual information prior to processing visual tokens, thereby facilitating more effective cross-modal alignment and reasoning. Such a fusion strategy significantly strengthens semantic contextualization, particularly in scenarios involving occlusion, low contrast, or cluttered backgrounds. By providing the model with explicit high-level semantics ahead of localized visual information, the proposed design improves the robustness, completeness, and discriminative capacity of small-object detection within the teacher model framework.

*2) Implicit Knowledge Alignment:* To address the feature misalignment and information redundancy that may arise during direct knowledge transfer from a multimodal teacher model, this study adopts a dual-level distillation strategy, feature layer distillation and detection head distillation, as illustrated in the redesigned Fig. 6 diagram of the proposed two-level knowledge distillation framework. The teacher model provides five multiscale feature maps, each aligned with the corresponding feature map of the student model through feature layer distillation. In parallel, the outputs of the teacher's classification and regression heads supervise the student's detection heads via detection head distillation. This mechanism ensures comprehensive and stable transfer of multimodal knowledge to the image-only student model. This design ensures stable and accurate cross-modal knowledge transfer while enabling the student model to internalize semantic information even without textual input at inference time.

1) The teacher model produces five multiscale feature maps from the Transformer encoder. Each feature map is individually aligned with the corresponding hidden representation of the student model at the same spatial resolution. This one-to-one multiscale alignment minimizes the feature discrepancy between the two models and implicitly transfers high-level semantic cues originating from the textual modality, thereby enhancing the student's representational capability in image-only settings.
2) Beyond feature-level alignment, we further distill the predictions of the teacher's classification and regression heads into the student model. By aligning the output distributions of both detection heads, the student acquires the teacher's decision priors, improves its discrimination of difficult samples, and reduces feature drift caused by modality differences.

The combination of these two components enables the student model to effectively leverage multimodal semantics during training while maintaining high efficiency in single-modal inference, as detailed in Fig. 6. Building upon the feature alignment strategy, we further introduce a two-stage training scheme to ensure robust single-modal inference.

*3) Two-Stage Multimodal Training and Single-Modal Inference Adaptation:* To leverage textual modality's semantic benefits during training while ensuring efficient image-only inference post-deployment, a two-stage training approach is proposed.

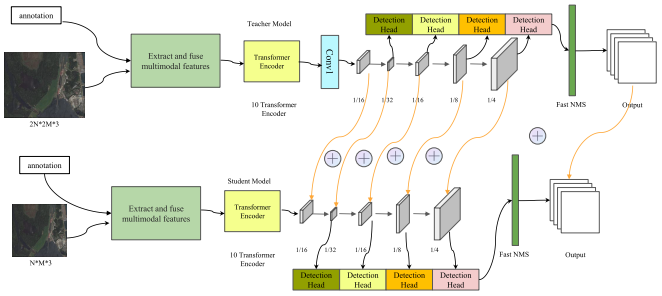



Fig. 6. Diagram of two-stage knowledge distillation and implicit knowledge alignment.

TABLE I
DATASET INFORMATION DESCRIPTION

| Dataset | Images | Instances | Classes | Annotation Type |
|---|---|---|---|---|
| DOTA-v1.0 | 2,806 | 188,282 | 15 | Oriented Bounding Box |
| DIOR-R | 23,463 | 192,472 | 20 | Oriented Bounding Box |
| VEDAI | 1,210 | 3,000+ | 12 | Horizontal Bounding Box |
| NWPU | 800 | 3,651 | 10 | Horizontal Bounding Box |

1) *Multimodal Student Model:* In the first stage of multimodal distillation, the student model learns from image–text inputs by aligning features and detection heads with the teacher. This allows the student to internalize semantic cues from text, preserving detection performance even without textual input at inference.
2) *Single-Modal Student Model:* For image-only deployment, the Transformer encoder weights are frozen while the feature pyramid and detection head undergo fine-tuning (as shown in Fig. 7). This phase employs image-based supervised learning to optimize frontend feature extraction and backend detection modules, enabling efficient single-modal inference by retaining cross-modal knowledge internalized during prior training.

## IV. EXPERIMENTAL RESULTS

### A. Experimental Setup

We conducted extensive experiments on four widely used AAV and remote sensing datasets to evaluate the effectiveness and generalizability of MDADet, as summarized in Table I. These datasets include DOTA-v1.0, DIOR-R, VEDAI, and NWPU, which collectively present diverse object categories, scene complexities, and annotation types.

*DOTA-v1.0 [34]:* This dataset contains 2806 high-resolution images with 188 282 object instances across 15 categories, annotated with oriented bounding boxes. It covers a wide variety of aerial scenes, including urban areas, airports, and industrial zones, providing a challenging benchmark for multiclass and densely packed object detection.

*DIOR-R [35]:* DIOR-R comprises 23 463 images with 192 472 object instances spanning 20 classes, also annotated with oriented bounding boxes. The dataset includes objects at different scales and orientations in complex natural and

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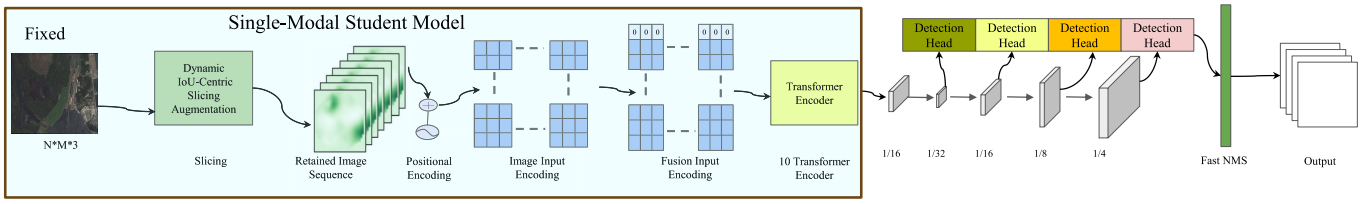



Fig. 7. Diagram of the overall fine-tuning framework.

TABLE II
TRAINING PARAMETERS

| Parameter | Configuration |
|---|---|
| Development Framework | PyTorch |
| GPU | NVIDIA RTX 4090 |
| Image size of teacher model | 1024×1024 |
| Image size of student model | 512×512 |
| Optimizer | Adam |
| Optimizer $\beta_1$ | 0.99 |
| Optimizer $\beta_2$ | 0.999 |
| Weight decay | 0.05 |
| Batch size | 2 |
| Epoch | 300 |

urban environments, making it suitable for evaluating model robustness against rotation variations and scale diversity.

*VEDAI [36]:* This dataset contains 1210 images with over 3000 annotated objects across 12 categories using horizontal bounding boxes. It primarily focuses on vehicles and transportation-related objects in aerial scenes. The dataset's relatively sparse distribution of objects tests a detector's ability to capture small and moderately sized targets under varying background conditions.

*NWPU [37]:* NWPU consists of 800 images with 3651 instances from ten classes, annotated with horizontal bounding boxes. The dataset features high-resolution aerial imagery with dense object distributions and diverse backgrounds, challenging detectors to identify small objects with low contrast or overlapping instances.

The proposed framework is implemented in PyTorch and evaluated on a workstation with an NVIDIA 4090 GPU using four datasets. Following [38], a tenfold cross-validation is performed, with nine folds for training and one for testing in each iteration. Results are reported as the average across all folds. Input images are resized to $1024 \times 1024$ for the teacher model and $512 \times 512$ for the student model. Training employs data augmentation techniques including HSV color transformation, multiscale training, translation, horizontal flipping, and mosaic augmentation, while no augmentation is applied during testing. The network is trained using the Adam optimizer [38] with parameters set to $\beta_1 = 0.9$ and $\beta_2 = 0.999$, a weight decay of 0.05, and a batch size of 2. The entire training process spans 300 epochs. The key hyperparameters and training details as shown in Table II. We selected RetainNet [39], YOLOV3 [39], [40], OR-CNN [39], FMSSD [39], [41], ViT-B [42], R3Det [43], FPN [44], LSKNet-S [45], and LEGNet [46] as comparison models.

As Table III shows, MDADet achieves the highest overall performance with an mAP of 81.07%, outperforming the best two-stage detector R3Det (77.36%) by 3.71% and the leading one-stage method LEGNet (79.37%) by 1.70%. These results underscore the effectiveness of MDADet in achieving a superior balance between detection accuracy and architectural efficiency. Compared with LSKNet-S, YOLO, ViT-B, and LEGNet, both of which exhibit noticeable missed detections in regions containing densely distributed small targets, MDADet demonstrates a markedly stronger ability to capture fine-grained object cues even under severe background interference. The improvement stems from MDADet's multimodal guidance and hierarchical feature enhancement, which together produce more discriminative feature representations for small-object perception. As shown in the visualization, MDADet consistently identifies all objects within cluttered environments, whereas baseline methods tend to ignore targets with low contrast or heavy overlap. These results further validate the robustness and reliability of MDADet when applied to complex real-world aerial scenes.

In addition, we provide an extended cross-dataset visualization analysis using representative samples from DOTA, DIOR-R, VEDAI, and NWPU. Since the textual modality in MDADet is generated directly from annotation-derived descriptions encoded by RoBERTa, the multimodal pipeline is inherently compatible with any dataset containing standard bounding-box labels, ensuring broad practical applicability. The results in Fig. 8 demonstrate that MDADet reliably detects vehicles under significant occlusion and arbitrary rotation, outperforming several compared models that exhibit clear miss detections. Fig. 9 highlights the robustness of our approach in dense small-object scenarios, where MDADet misses only two instances compared with four to nine missed detections by competing methods. Fig. 10 shows that MDADet effectively identifies low-visibility targets under missing information and extreme viewing angles, whereas models such as YOLOv3, Ghost, RetainNet, and SuperYOLO fail to capture these objects. Fig. 11 further indicates that in scenes with medium-scale, sparse objects, all methods perform consistently, illustrating that our advantages are especially significant under high density, multiscale variation, or orientation complexity. These observations collectively verify that MDADet scales effectively across diverse aerial imaging conditions and maintains strong generalization across datasets with different object distributions and geometric characteristics.

Table IV presents the extended comparative evaluation of MDADet against a comprehensive set of representative two-stage and one-stage detectors, including additional baselines YOLO11 [47], GHOST [48], SuperYOLO [49], and Mamba [50]. The experiments are conducted across three remote sensing datasets: VEDAI, DIOR-R, and NWPU. MDADet achieves the highest mAP of 86.76% on VEDAI, 73.55% on DIOR-R,

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TABLE III

COMPARISON RESULTS OF DIFFERENT ALGORITHMS ON THE DOTA DATASET. BOLD AND UNDERLINE INDICATE THE BEST AND THE SECOND-BEST RESULTS, RESPECTIVELY

| Method | | Plane | BD | Bridge | GTF | SV | LV | Ship | TC | BC | ST | SBF | RA | Harbor | SP | HC | mAP |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Two Stage | FMSSD | 89.11 | 81.51 | 48.22 | 67.94 | 69.23 | 73.56 | 76.87 | 90.71 | 82.67 | 73.33 | 52.65 | 67.52 | 72.37 | 80.57 | 60.15 | 72.43 |
| | R3Det | 88.90 | 84.17 | 55.80 | 69.35 | 78.72 | 84.08 | 87.00 | 89.75 | 84.32 | 85.73 | 64.74 | 61.80 | 76.62 | 78.49 | 70.89 | 77.36 |
| | OR-CNN | 89.46 | 82.12 | 54.78 | 70.86 | 78.93 | 83.00 | 88.20 | 90.90 | 87.50 | 84.68 | 63.97 | 67.69 | 74.94 | 68.84 | 52.28 | 75.87 |
| | FPN-RCNN | 88.70 | 75.10 | 52.60 | 59.20 | 69.40 | 78.80 | 84.50 | 90.60 | 81.30 | 82.60 | 52.50 | 62.10 | 76.70 | 66.30 | 60.10 | 72.27 |
| One Stage | RetainNet | 78.22 | 53.41 | 26.38 | 42.27 | 63.64 | 52.63 | 73.19 | 87.17 | 44.64 | 57.99 | 18.03 | 51.00 | 43.39 | 56.56 | 47.44 | 50.39 |
| | YOLOV3 | 79.00 | 77.10 | 33.90 | 68.10 | 52.80 | 52.20 | 49.80 | 89.90 | 74.80 | 59.20 | 55.50 | 49.00 | 61.50 | 55.90 | 41.70 | 60.00 |
| | LSKNet-S | 89.66 | 85.52 | 57.72 | 75.70 | 74.95 | 78.69 | 88.24 | 90.88 | 86.79 | 86.38 | 66.92 | 63.77 | 77.77 | 74.47 | 64.82 | 77.49 |
| | ViT-B | 89.33 | 79.82 | 54.99 | 70.83 | 79.89 | 85.25 | 88.01 | 90.88 | 83.83 | 86.23 | 56.30 | 68.19 | 75.98 | 76.42 | 69.93 | 77.05 |
| | LEGNet | 89.22 | 85.04 | 55.59 | 75.73 | **81.42** | 84.56 | 88.32 | 90.90 | 87.90 | 87.02 | 65.98 | 70.13 | 77.56 | **81.88** | 69.26 | 79.37 |
| | MDADet | **89.84** | **86.92** | **61.13** | **77.52** | 81.24 | **86.37** | **89.55** | **91.40** | **88.91** | **88.74** | **69.25** | **71.60** | **79.80** | 80.32 | **74.11** | **81.07** |

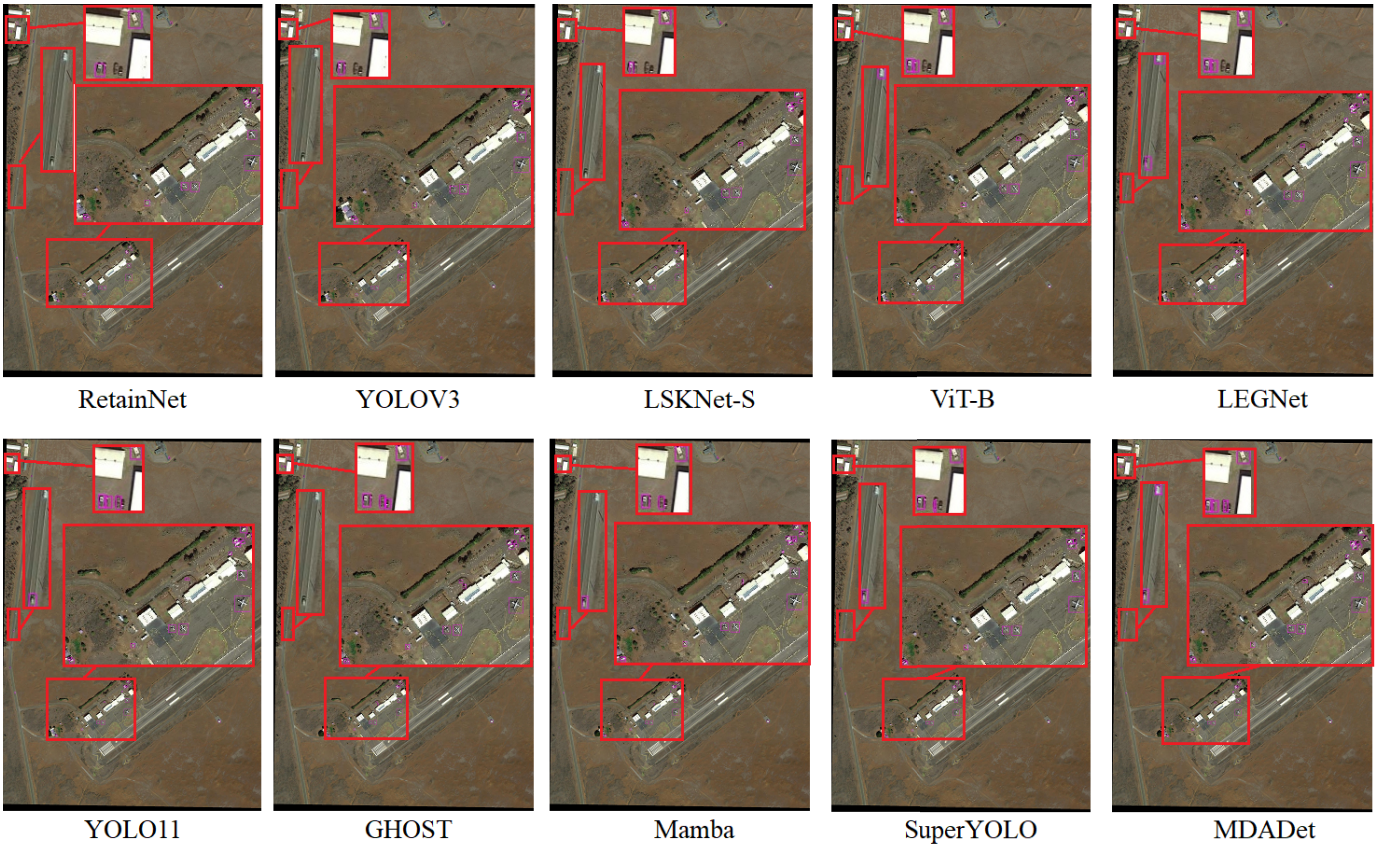



Fig. 8. Visualization of detection results of DOTA.

and 97.61% on NWPU VHR-10, surpassing the strongest baselines by notable margins. Visualizations in Figs. 9–11 further demonstrate that MDADet accurately captures target regions, particularly in dense small-object and cluttered background scenarios. To account for practical efficiency, we report computational cost in terms of model parameters, FLOPs, and FPS. MDADet maintains a competitive inference speed while requiring only 37.8-M parameters and 147.63-G FLOPs on average, significantly lower than larger baselines such as ViT-B and R3Det. This combination of high accuracy and efficient computation highlights MDADet's strong contribution in small-object detection tasks and confirms its applicability to real-world aerial imagery scenarios with diverse object scales, scene densities, and environmental variations.

As reported in Table IV, MDADet achieves a real-time inference speed of 29.4 FPS on an NVIDIA 4090, which satisfies the typical performance requirements for AAV-based on-board detection systems. In comparison, SuperYOLO and Mamba achieve higher speeds at 45 and 43.1 FPS, but with noticeably lower detection accuracy across VEDAI, DIOR-R, and NWPU. This observation indicates that MDADet maintains a more favorable accuracy efficiency balance.

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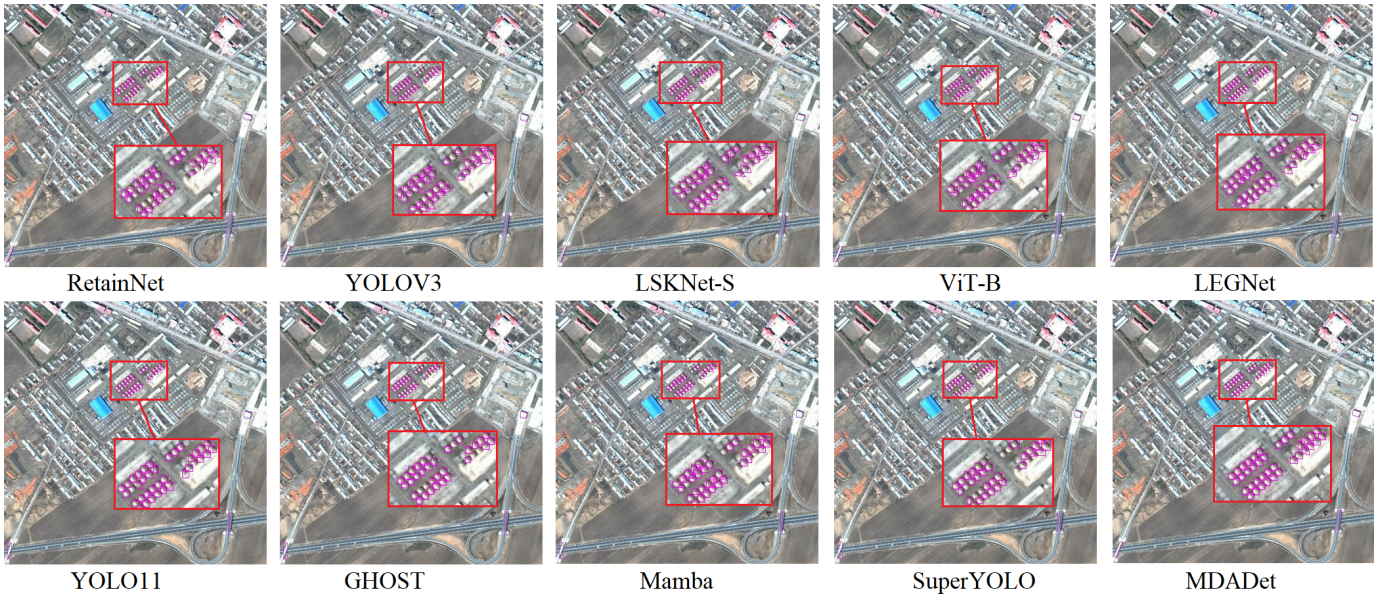



Fig. 9. Visualization of detection results of DIOR.




Fig. 10. Visualization of detection results of VEDAI.

### B. Ablation Study

Ablation studies on the DOTA dataset are conducted to rigorously assess the impact of each proposed strategy, as outlined in Sections IV-B1–IV-B3.

*1) Ablation Study on Textual Representation Quality:* To investigate the impact of textual representation quality on detection performance, we designed four ablation settings for the teacher model.

1) *Target-Only Description (a):* Only the bounding box target is described in text.
2) *Target + Background Description (b):* Both target and surrounding background are described.
3) *Target-Only Description With 10% Noise (c):* Random noise is injected into 10% of words in the target description.
4) *Target + Background Description With 10% Noise (d):* Random noise is added to 10% of words in the combined target and background description.

Table V results show that Setting (a), using pure target descriptions, achieves the highest mAP of 81.62%, demonstrating the most effective guidance for cross-modal knowledge distillation. Incorporating background information (b) slightly lowers performance to 80.35%, as irrelevant textual content increases the transformer's reliance on text without providing additional useful semantics. Introducing 10% noise into the target-only descriptions (c) results in a similar mAP of 81.10%, indicating that moderate noise has a limited impact when the textual information is otherwise precise. The combination of background information with noise (d) produces the lowest mAP of 80.04%, highlighting the

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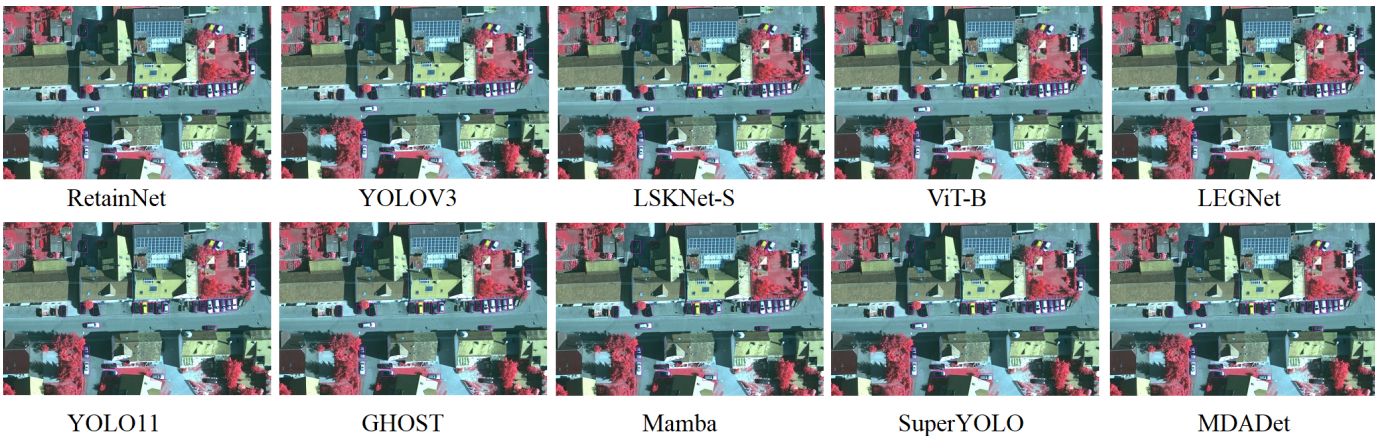



Fig. 11. Visualization of detection results of NWPU.

TABLE IV

COMPARISON RESULTS OF DIFFERENT ALGORITHMS ON THE THREE DATASETS. BOLD AND UNDERLINE INDICATE THE BEST AND THE SECOND-BEST RESULTS, RESPECTIVELY

| Method | | Dataset | | | Mean Params (M) | Mean FLOPs (G) | Speed (Fps) |
|---|---|---|---|---|---|---|---|
| | | VEDAI | DIOR | NWPU | | | |
| Two Stage | FMSSD | — | 69.50 | 90.40 | 136.04 | — | 30.20 |
| | R3Det | 76.35 | 68.30 | 87.64 | 41.90 | 336.00 | — |
| | OR-CNN | 81.10 | 64.30 | 84.40 | 41.17 | 331.00 | 21.50 |
| | FPN-RCNN | 77.67 | 47.84 | 80.90 | 42.70 | 239.30 | — |
| One Stage | RetainNet | 77.82 | 65.70 | 88.60 | 55.39 | 293.36 | 28.60 |
| | YOLOV3 | 61.06 | 57.10 | 90.53 | 61.63 | 198.92 | 31.80 |
| | LSKNet-S | 82.82 | 65.90 | 92.47 | 36.62 | 194.61 | 22.50 |
| | ViT-B | 80.88 | 66.65 | 94.25 | 112.47 | 364.56 | — |
| | LEGNet | 83.04 | 68.40 | <u>96.21</u> | 23.90 | <u>132.85</u> | 20.90 |
| | YOLO11 | <u>84.30</u> | 65.60 | 85.01 | 58.87 | 519.00 | 20.70 |
| | GHOST | 80.31 | 71.53 | 91.97 | 19.30 | 265.00 | — |
| | Mamba | 81.20 | — | — | <u>17.07</u> | — | <u>43.10</u> |
| | SuperYOLO | 75.90 | <u>71.82</u> | 93.30 | **7.09** | **54.13** | **45.00** |
| | MDADet | **86.76** | **73.55** | **97.61** | 37.80 | 147.63 | 29.40 |

TABLE V

ABLATION STUDY RESULTS OF TEXTUAL QUALITY

| Setting | a | b | c | d |
|---|---|---|---|---|
| mAP (%) | 81.62 | 80.35 | 81.10 | 80.04 |

negative effect of both redundancy and noise in textual embeddings. Overall, these findings indicate that high-quality, noise-free target descriptions are critical for effective multimodal feature alignment, as redundant or noisy text can interfere with the student model's ability to retain semantic knowledge and maintain robust image-only inference.

*2) Effectiveness of Multimodal Input and Distillation:* To evaluate the effectiveness of the proposed multimodal inputs and distillation framework, we conduct a detailed ablation study on the DOTA dataset. The experiment compares the performance of a multimodal student model (image + text input) and a single-modal student model (image-only input) under both knowledge distillation and nondistillation scenarios. The comparison is made across three key aspects: detection performance, model parameters, and computational cost. The results are summarized in Table VI.

The experiments now separately evaluate feature layer distillation, detection head distillation, their combination (with distillation), and the no-distillation baseline, for both multimodal student model (image + text input) and single-modal student model (image-only input after fine-tuning). The ablation experiments are designed as follows.

1) *Teacher Model:* Trained using both image and text inputs, serving as a high-performance baseline (mAP 81.62%).
2) *Feature Layer Distillation:* Only the intermediate feature layers are distilled from the teacher to the student. The multimodal student model uses image + text inputs during distillation, while the single-modal student model is fine-tuned with image-only input to ensure effective single-modal performance.
3) *Detection Head Distillation:* Only the detection head is distilled. The multimodal student model uses image + text, and the single-modal student model is fine-tuned with image-only input.
4) *With Distillation:* Both feature layers and the detection head are distilled together. The multimodal student model uses image + text during distillation, and the single-modal student model is fine-tuned with image-only input.
5) *Without Distillation:* No distillation is performed; the student is trained directly with image + text inputs.

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TABLE VI
DISTILLATION ABLATION EXPERIMENTAL RESULTS

| Method | | Input | mAP(%) | Params(M) | FLOPs(G) |
|---|---|---|---|---|---|
| Teacher Model | | Image+Text | 81.62 | 141.8 | 415.4 |
| Feature Layer Distillation | Multi-Modal Student Model | Image+Text | 80.85 | 52.9 | 195.7 |
| | Single-Modal Student Model | Only Image | 80.72 | 37.8 | 147.6 |
| Detection Head Distillation | Multi-Modal Student Model | Image+Text | 79.24 | 52.9 | 195.7 |
| | Single-Modal Student Model | Only Image | 78.93 | 37.8 | 147.6 |
| With Distillation | Multi-Modal Student Model | Image+Text | 81.17 | 52.9 | 195.7 |
| | Single-Modal Student Model | Only Image | 81.07 | 37.8 | 147.6 |
| Without Distillation | Multi-Modal Student Model | Image+Text | 69.53 | 52.9 | 195.7 |
| | Single-Modal Student Model | Only Image | 64.86 | 37.8 | 147.6 |

TABLE VII
ABLATION STUDY RESULTS OF KEY COMPONENTS

| Method | Input | Adaptive data augmentation | Dynamic anchors | Multi-task loss optimization | mAP (%) | Params (M) | FLOPs (G) |
|---|---|---|---|---|---|---|---|
| Single-Modal Student Model | Only Image | ✓ | ✓ | ✓ | 81.07 | 37.8 | 147.6 |
| | | × | ✓ | ✓ | 78.14 | 37.8 | 231.8 |
| | | ✓ | × | ✓ | 78.63 | 37.8 | 183.5 |
| | | ✓ | ✓ | × | 76.89 | 37.8 | 147.6 |

The single-modal student model is fine-tuned with image-only input.

a) The teacher model achieves 81.62% mAP using multimodal inputs, serving as a strong baseline. With distillation, the multimodal student retains 99.45% of this performance while reducing parameters by 62.7% and FLOPs by 52.9%, whereas the single-modal student attains 81.07% mAP with 73.3% fewer parameters and 64.5% fewer FLOPs compared to the teacher. Notably, without distillation, student models experience severe performance degradation (12.09% and 16.76% mAP drops), underscoring the necessity of cross-modal knowledge transfer for effective model compression and adaptation.

b) Distillation improves mAP by 11.64% (69.53%–81.17%) through teacher–student alignment of intermediate features and detection heads. Feature layer distillation alone allows the multimodal student model to achieve 80.85% mAP and the single-modal student model 80.72% mAP, demonstrating the effectiveness of intermediate feature guidance. Detection head distillation alone yields 79.24% and 78.93% mAP for multimodal and single-modal students, respectively. When both feature and head distillation are combined, the single-modal student model reaches 81.07% mAP, closely approaching the teacher's performance with substantially reduced parameters and FLOPs. Without any distillation, performance drops dramatically to 69.53% for multimodal and 64.86% for single-modal students, confirming the essential role of cross-modal knowledge transfer.

c) Without distillation, the performance gap widens to 4.67% mAP (69.53% versus 64.86%), confirming that textual cues provide critical supervisory signals during training, particularly for small/occluded objects.

*3) Ablation Studies on Key Components:* We conducted comprehensive ablation experiments on the DOTA dataset using the single-modal student model to evaluate the effectiveness of three critical components: adaptive data augmentation, dynamic anchors, and multitask loss optimization. Detailed quantitative results are summarized in Table VII.

1) The adaptive data augmentation strategy substantially enhanced model performance, achieving a 2.93% improvement in mAP from 78.14% to 81.07%. Concurrently, computational complexity was reduced by approximately 36.3%, as evidenced by a decrease in FLOPs from 231.8 to 147.6 G.
2) Dynamic anchors also proved essential, yielding a 2.44% mAP improvement from 78.63% to 81.07% without increasing the parameter count, while lowering FLOPs from 183.5 to 147.6 G. This confirms that dynamic anchor design enhances scale-feature alignment and reduces redundant computations, which is crucial for real-time AAV detection systems.
3) Multitask loss optimization yielded the most substantial gain, improving mAP by 4.18% from 76.89% to 81.07% without affecting computational load. The task-aware loss with dynamic weight balancing effectively addresses the complex gradient coordination needed for multiscale and densely distributed aerial objects, which surpasses traditional loss functions.

Collectively, these ablation results highlight the key importance of adaptive augmentation, dynamic anchor mechanisms, and optimized multitask loss in improving reliable and efficient aerial object detection under practical conditions.

## V. CONCLUSION

This work presents MDADet, a multimodal dynamic adaptive framework tailored for small-object detection in AAV

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imagery. The main contributions of this study are summarized as follows.

1) *Multimodal Fusion and Cross-Modal Adaptation Architecture:* MDADet designs a multimodal teacher model that fuses high-resolution aerial images with RoBERTa-encoded positional descriptions. Cross-modal knowledge is distilled to the student model through feature alignment, contrastive representation learning, and detection head supervision. A two-stage training scheme ensures the student model retains semantic gains from multimodal training while enabling robust image-only inference.
2) *Dynamic Data Augmentation and Multiscale Composite Loss Design:* To mitigate background redundancy and the sparsity of small objects, MDADet incorporates DICSA to prioritize informative regions and a multi-scale optimized composite loss to enhance localization and classification of small, marginal, and low-contrast objects.
3) *Lightweight Multiscale Feature Fusion:* MDADet employs convolutional layers and pixel shuffle for efficient multiscale feature generation, combined with hierarchical task allocation and Fast NMS to improve both detection speed and accuracy.

Extensive experiments on multiple AAV and remote sensing benchmarks demonstrate that MDADet achieves competitive or superior performance on multiple benchmarks, particularly excelling in small-object detection scenarios. These results validate the robustness, efficiency, and generalizability of the proposed method for real-world aerial detection scenarios.

For future research, we will focus on explorations in two directions. On one hand, we will continue to optimize the algorithm's theoretical aspects, attempting to explore weakly supervised learning and edge computing optimization to expand its applicability in unsupervised aerial scenarios [51]. On the other hand, from the perspective of algorithm application, we will try to explore the inference performance and speed of this algorithm on real AAV hardware platforms. Furthermore, inspired by recent advances in super-resolution assisted detection, such as the hybrid deep learning approach proposed in [52] and [53], we plan to investigate integrating enhanced super-resolution techniques with MDADet. While our current pixel-shuffle-based up-sampling efficiently generates multiscale high-resolution features with reduced computational cost, incorporating super-resolution methods may further improve the visibility and detection accuracy of small and densely distributed targets, representing a promising direction for future work.

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